Renxiang Chu

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EDUCATION

Peking University **Beijing, China**

- Bachelor of Medicine in Basic Medical Sciences *Expected 2027*
  Dual Degree in Economics

Core Courses:

Medical Molecular and Cell Genetics, Human Morphology and Functional Regulation, Nervous System, Pathogenic Biology and Medical Immunology, Probability and Statistics, Intermediate Microeconomics, Intermediate Macroeconomics

RESEARCH EXPERIENCE

FundusAge: A Deep Learning Approach to Physiological Aging Estimation and Metabolic Biomarker Discovery

Position: Research Assistant **Supervisor:** *Meng Lu* *Sept. 2024 - Present*

- **Publication:** Chu, R.*, Sun, A.*, Qu, J., & Lu, M. (2026). ***Disentangling physiological heterogeneity in retinal aging using a deep learning-based biological age framework.*** Preprint at medRxiv. https://doi.org/10.64898/2026.02.13.26346265
  (* Co-first authors)
- Foundation Model Engineering: Developed a retinal aging clock by leveraging VisionFM, a ViT-based Masked Autoencoder (MAE) backbone pre-trained on 3.4 million ophthalmic images. Fine-tuned the model on a curated dataset of 112,019 fundus images, achieving state-of-the-art performance (MAE: 2.49 years) and robust generalization across diverse demographics.
- Latent Space Disentanglement: Designed a Teacher-Student Residual Learning Framework to explicitly disentangle physiological aging features from pathological noise in the latent space. This approach effectively isolated the physiological aging signal, enabling precise biological age estimation even in patients with ocular comorbidities.
- Multimodal Fusion & Interpretation: Constructed a Tabular Transformer using masked prediction and attention heatmaps to analyze interactions between biological age and clinical indicators, establishing physiological age as an independent biomarker.
- Applied SHAP values and feature importance analysis to quantify driving factors of age gaps, identifying inflammation markers as primary contributors to accelerated aging while linking hemodynamic pressures to physiological deviations.

ACADEMIC PROJECTS

Computational Characterization of Biomolecular Condensate Dynamics

- Developed customized data processing scripts in R and Python to analyze the compositional complexity and phase-separation dynamics of biomolecular condensates, enabling the quantitative characterization of protein-RNA interactions within complex cellular subsystems.
- Evaluated the spatiotemporal regulation of liquid-liquid phase separation through literature synthesis and experimental validation, integrating computational findings with biochemical assays to understand the structural stability and functional roles of macromolecular assemblies.

Knowledge-Augmented LLM for Precision Clinical Decision Support

- Fine-tuned a Llama-3-70B model using Parameter-Efficient Fine-Tuning and LoRA to enhance reasoning, implementing a Retrieval-Augmented Generation pipeline to anchor outputs in medical literature and evidence-based guidelines.
- Built a prompt-tuning strategy to extract diagnostic insights from unstructured clinical notes, improving the accuracy of symptom-disease mapping and reducing medical hallucinations through benchmarking against expert-curated health records.

LEADERSHIP & ACTIVITIES

Lead Vocalist, Peking University Health Science Center A Cappella Club

- Performed as a featured vocalist in the Peking University 2025 New Year Gala, delivering professional musical performances for a university-wide audience of thousands while maintaining high artistic and technical standards.
- Organized and led performances for the School of Basic Medical Sciences 2026 New Year's Eve Gala, fostering a vibrant community culture and balancing rigorous medical training with intensive creative rehearsals.

Team Leader, Peking University School of Basic Medical Sciences Social Practice Group

- Led a social practice initiative in Suzhou themed on industrial innovation, where I managed all organizational logistics and coordinated site-specific research activities for a delegation of Peking University medical students.
- Explored regional pharmaceutical ecosystems through field visits to Soochow University and Kunshan Biomedical Industrial Park, engaging with experts at Synermore Biologic to gain insights into the commercialization of innovative biotechnology.

SKILLS

- **Programming Languages:** *Python, R*
- **Developer Tools:** *Linux, AWS, Conda, Jupyter, NumPy, Pandas, LaTeX, Git, Markdown*
- **Languages:** *English, Mandarin Chinese*